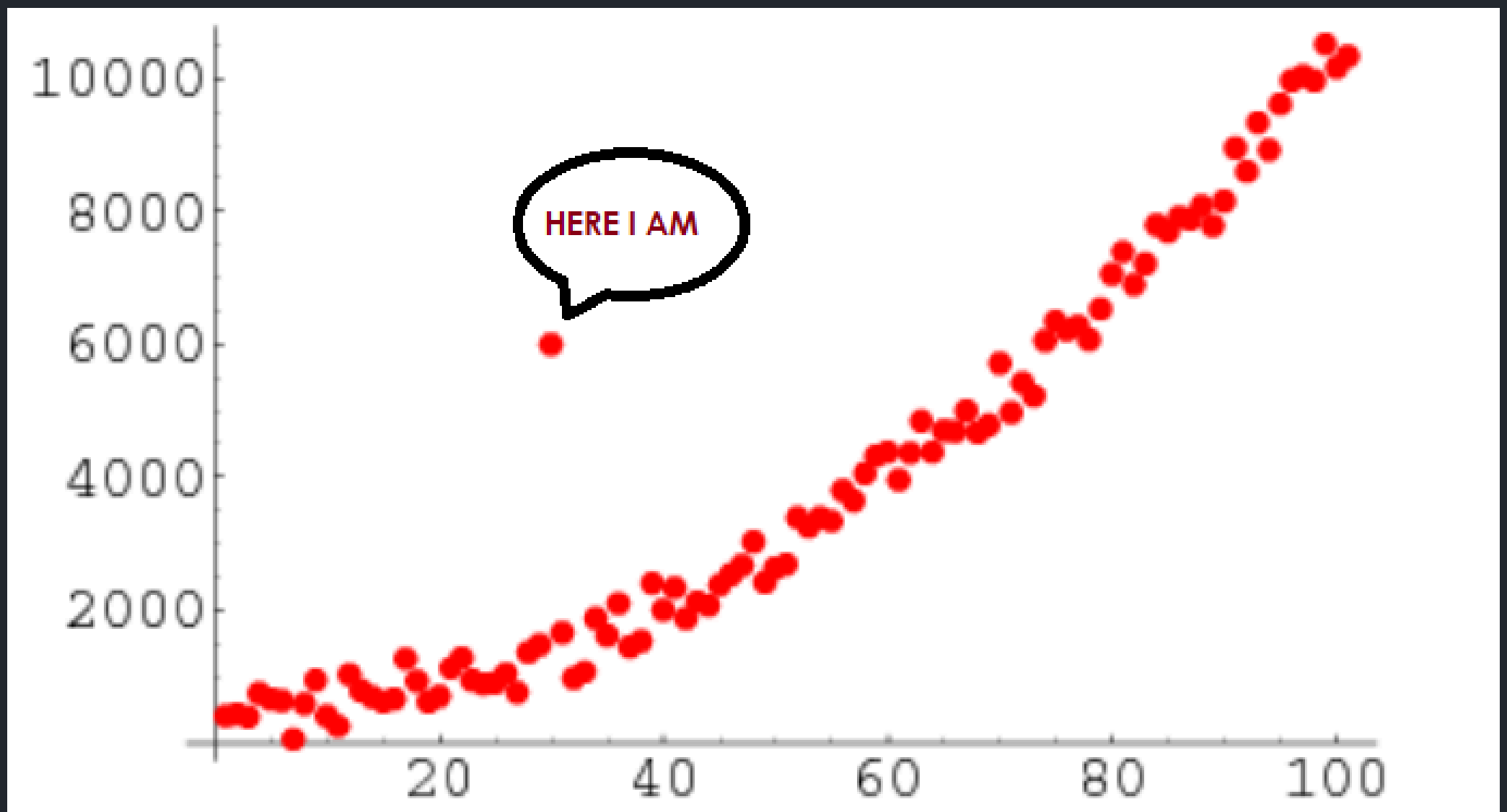
4 WAYS TO DEAL WITH OUTLIERS





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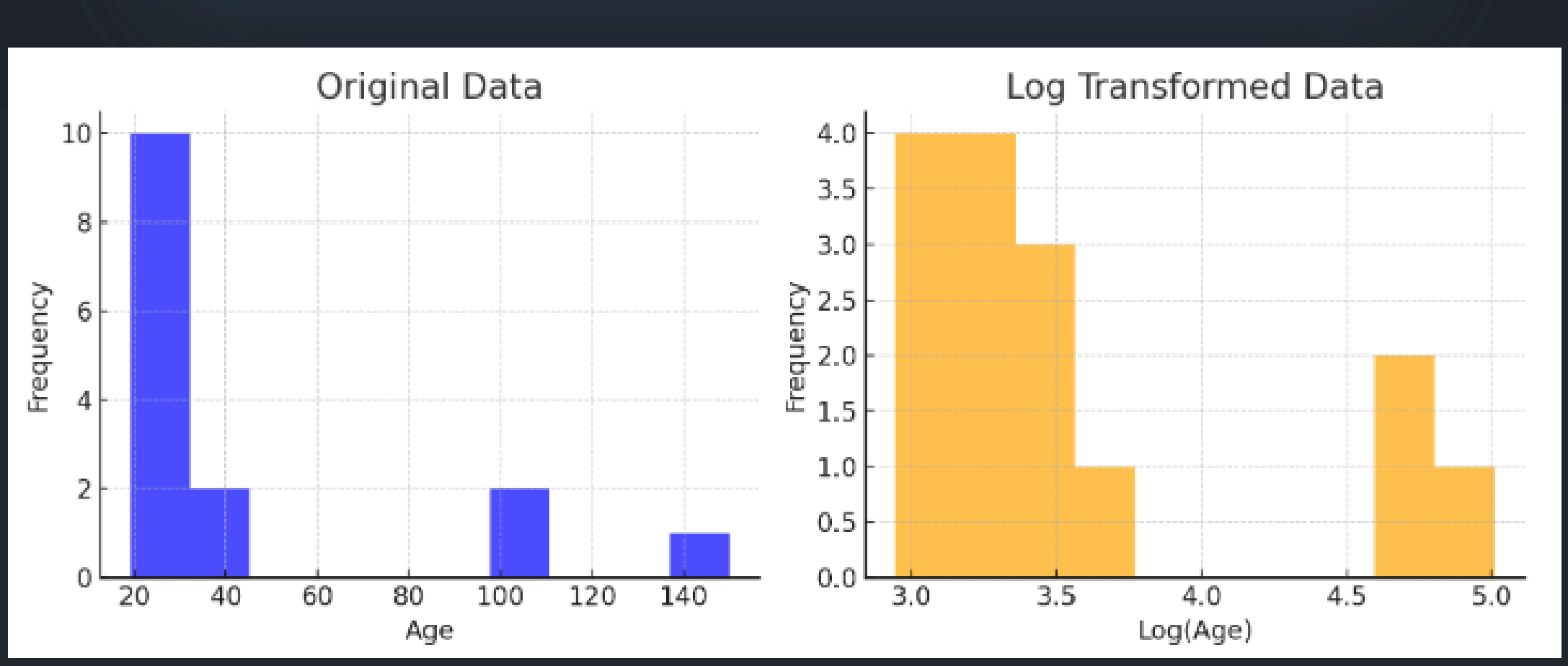
Transform

Transforming data using mathematical functions can help normalize the distribution and reduce the effect of outliers.

```
# Creating the dataset
data = pd.DataFrame({
    "Age": [25, 28, 29, 22, 30, 105,
    27, 21, 23, 150, 26, 19, 34, 38, 101]
})

# 1. Transform them: Log Transformation

data['Transformed_Age'] = np.log(data['Age'])
```



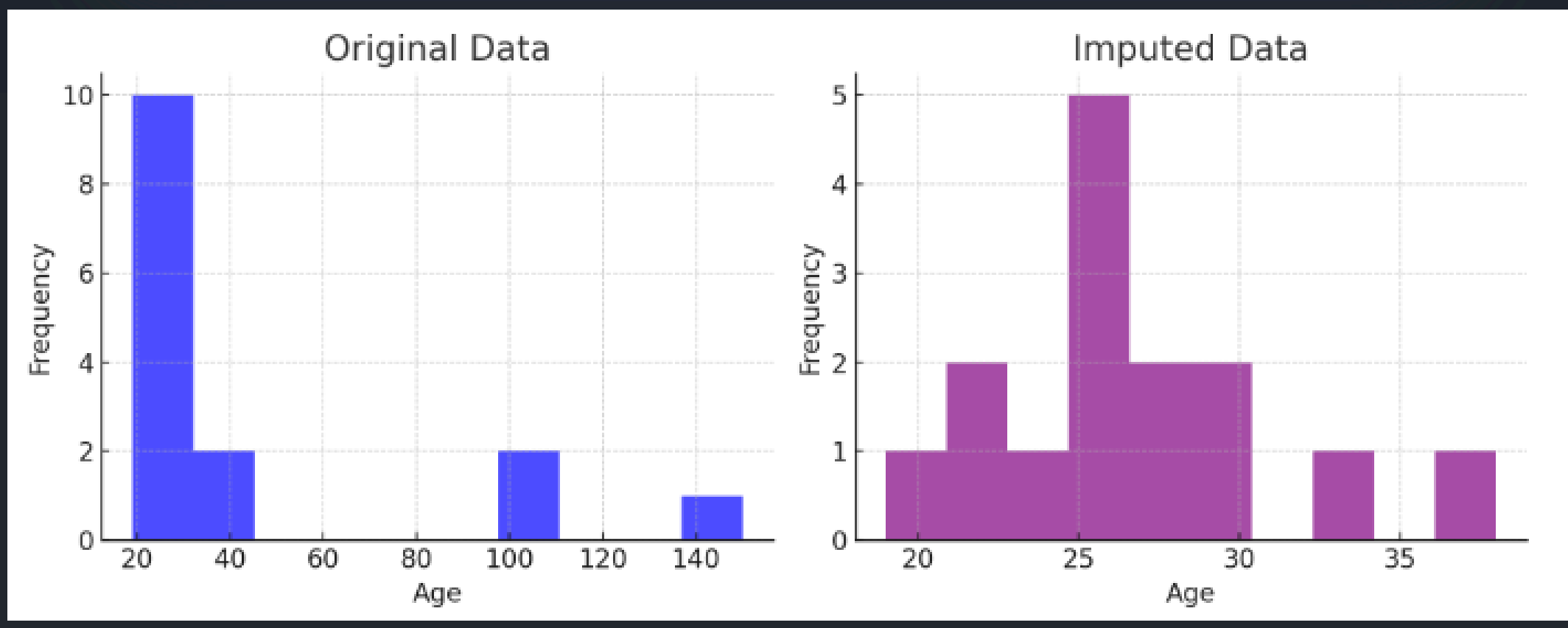
Impute

Replace outliers with a more typical value, often based on the central tendency (e.g., median)

```
# Creating the dataset
data = pd.DataFrame({
    "Age": [25, 28, 29, 22, 30, 105,
    27, 21, 23, 150, 26, 19, 34, 38, 101]
})

# 2. Impute them: Replacing outliers above age
# 100 with the median age

median_age = data[data['Age'] ⩽ 100]['Age'].median()
data['Imputed_Age'] = data['Age'].apply(lambda x:
median_age if x > 100 else x)
```



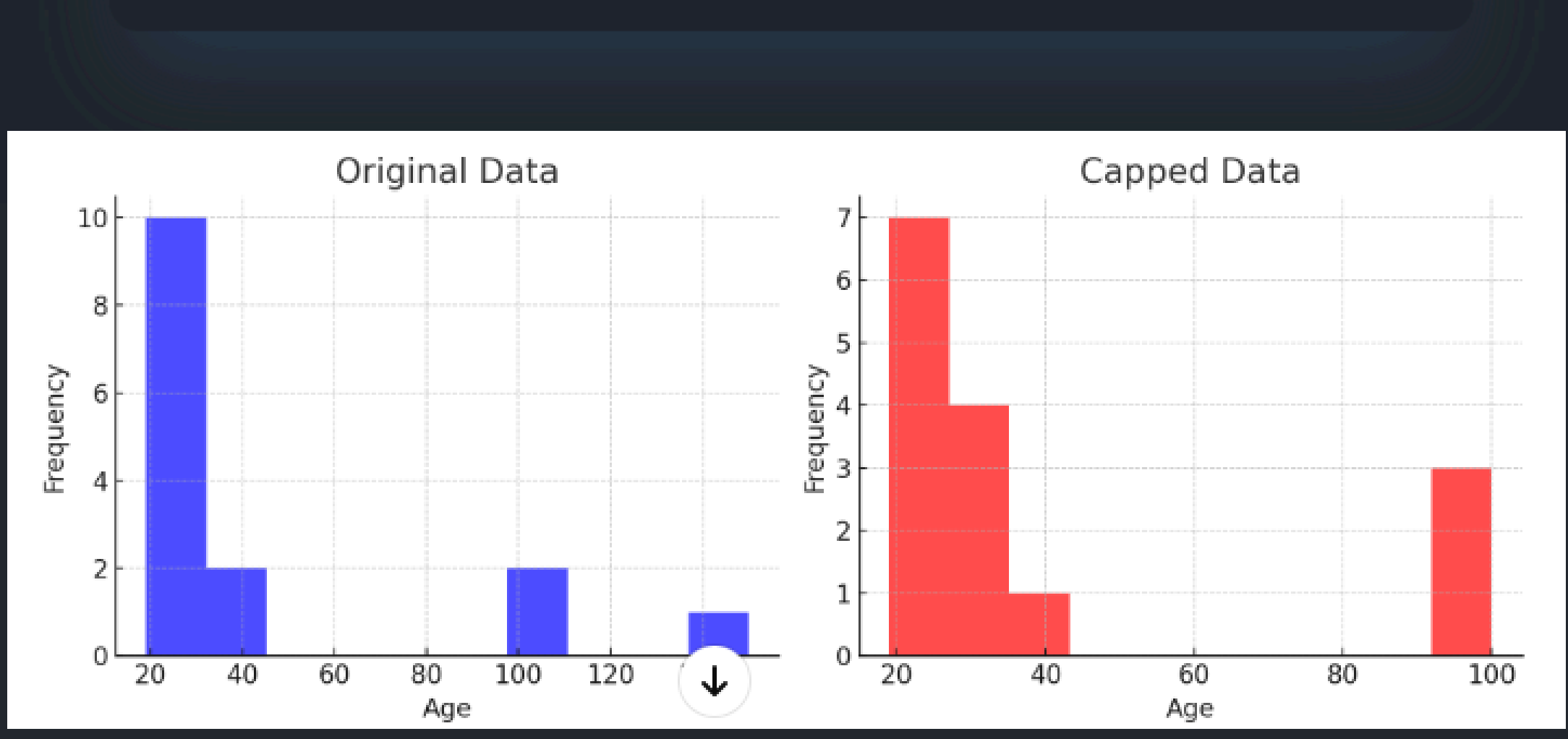
Cap

Limit the maximum or minimum value an outlier can have to reduce its effect.

```python
# Creating the dataset
data = pd.DataFrame({
    "Age": [25, 28, 29, 22, 30, 105,
    27, 21, 23, 150, 26, 19, 34, 38, 101]
})

# 3. Cap them: Capping age values above 100

max_age = 100
data['Capped_Age'] = data['Age'].apply(lambda x:
min(x, max_age))
```



Remove

Remove outliers by filtering them out based on a specific condition

```python
# Creating the dataset
data = pd.DataFrame({
    "Age": [25, 28, 29, 22, 30, 105,
    27, 21, 23, 150, 26, 19, 34, 38, 101]
})

# 4. Remove them: Removing outliers above age 100

cleaned_data = data[data['Age'] <= 100]
```

